HIGH-BANDWIDTH, LOW-LATENCY DATA TRANSMISSION SYSTEM AND PROTOCOL

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Title: HIGH-BANDWIDTH, LOW-LATENCY OPTICAL DATA CABLE AND TRANSMISSION PROTOCOL

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Abstract

A system and method for ultra-high-bandwidth data transmission over short distances. The system comprises a transmitter unit containing a high-density array of individually addressable photon emitters and a receiver unit containing a corresponding array of photon sensors. The transmitter unit encodes digital data into a rapid sequence of frames, where each emitter's wavelength and intensity represents a data packet. The receiver unit captures these frames and decodes them back into digital data. The massive parallelism of millions of emitter/sensor pairs allows for data transfer rates several orders of magnitude greater than conventional electrical or fiber-optic cables, with the total bandwidth being a direct function of the number of elements in the arrays. While a preferred embodiment uses a micro-display and an image sensor operating in the visible light spectrum, the principle is extendable to non-visible spectra such as infrared or ultraviolet. An integrated-circuit-based negotiation protocol allows the receiver unit to calibrate itself to the specific characteristics of the transmitter array, optimizing precision and data fidelity.

Background of the Invention

The field of data transmission is constantly driven by the demand for higher bandwidth and lower latency. In environments such as large-scale datacenters, high-performance computing (HPC) clusters, and emerging fields like quantum computing, the speed of data transfer between processing units is often the primary bottleneck limiting overall performance.

Conventional data transmission standards, such as CAT6 Ethernet, are limited in their data-carrying capacity by physical constraints related to electromagnetic interference and signal degradation over distance. While fiber optic cables offer significantly higher bandwidth, their use in short-range, high-density interconnects can be complex and costly. Furthermore, both technologies transmit data serially, or over a limited number of parallel channels.

In specialized fields like quantum computing, the computational power of the processing units has advanced at a rate that far outpaces the development of interconnect technologies capable of handling the vast amounts of data generated. There exists a critical need for a novel, short-range data transmission medium that offers a step-change increase in bandwidth, moving beyond the limitations of traditional serial data streams. The present invention addresses this need by utilizing the entire visible light spectrum across a massively parallel array of pixels as a medium for data transfer.

Summary of the Invention

The present invention provides a system and method for short-range, ultra-high-bandwidth data transmission. The core of the invention is a hardware interface that uses an array of individually addressable photon emitters as a transmitter and a corresponding array of photon sensors as a receiver.

Data is not transmitted as a single serial stream, but rather as a sequence of frames. Each element in the emitter array acts as an independent data channel, and its characteristics (e.g., wavelength, intensity) encode a packet of data. By transmitting millions of channels in parallel, the system achieves a total data bandwidth that is a direct function of the number of elements in the array (i.e., the resolution). This parallelism results in transfer speeds orders of magnitude beyond current technologies.

The principles of the invention are applicable across the electromagnetic spectrum. A preferred embodiment, which is easy to manufacture with current technology, utilizes a compact, high-resolution screen (such as a micro-OLED) as the photon emitter array and a compact, high-resolution camera sensor (such as a CMOS sensor) as the photon sensor array, operating within the visible light spectrum. In this embodiment, a data packet is encoded

in the color and brightness of each pixel.

A key feature of the invention is a built-in negotiation and calibration protocol, managed by integrated circuits within the interface connectors. This allows the receiver to query the transmitter for its specific capabilities—such as resolution, available wavelengths (color depth), and color matrix—and to adapt its decoding algorithm on-the-fly to maximize accuracy and noise rejection. Several physical embodiments are disclosed, making the technology adaptable to different applications from commercial datacenters to specialized scientific instruments.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

Brief Description of the Drawings

FIG. 1 is an isometric view of an embodiment of the assembled optical data cable, showing the two interlocking connector heads.

FIG. 2 is an exploded view of a connector head, illustrating the internal arrangement of the transmitter (TX) display or the receiver (RX) image sensor.

FIG. 3 is a block diagram illustrating the data flow path from a host computing device to a receiving computing device through the optical data cable system.

FIG. 4 is a conceptual diagram illustrating a "channel pixel" composed of a grid of multiple physical pixels, a technique used in one embodiment to enhance robustness.

FIG. 5 is a diagram illustrating three potential physical coupling configurations between the transmitter display and the receiver sensor.

Detailed Description of the Invention

The present invention provides a novel apparatus and method for ultra-high-bandwidth data transmission. The core principle is the use of a massively parallel array of electromagnetic radiation emitters and a corresponding array of sensors to transmit data. While the following description details a preferred embodiment operating in the visible light spectrum (referred to as the Kinescopic Optical Link (KOL) or Camera-Coupled Pixel (CCP) system), it is to be understood that the fundamental principles apply equally to non-visible spectra, such as infrared (IR) and ultraviolet (UV), by selecting emitter and sensor materials appropriate for those wavelengths.

Referring to **FIG. 1**, the system takes the physical form of a cable (100) with two interlocking connector heads (110a, 110b). The housing of the connectors is designed to ensure precise physical alignment and to block ambient radiation, creating an isolated channel for data transmission.

As shown in the exploded view of **FIG. 2**, each connector head contains a key electronic component. The transmitter (TX) connector head (110a) houses a compact, high-density photon emitter array. In the preferred embodiment, this is a micro-OLED display (210). The receiver (RX) connector head (110b) houses a compact, high-sensitivity photon sensor array, which in the preferred embodiment is a CMOS image sensor (220). Each connector head also contains an integrated circuit (IC) (230) responsible for managing the encoding/decoding logic and the inter-connector negotiation protocol.

The fundamental principle of operation is illustrated in the block diagram of **FIG. 3**. A host computing device (301) sends digital data to the system. The IC (230) in the TX connector head (110a) functions as an encoder. It translates the data stream into a sequence of two-dimensional frames. In each frame, a packet of data is mapped to the emission characteristics (e.g., wavelength and intensity, which in the visible spectrum correspond to color and brightness) and spatial position of an emitter in the array (210). The total data bandwidth of the system is a direct function of the number of emitters in the array (i.e., its resolution) and the refresh rate of the frames.

The emitter array (210) emits photons corresponding to these frames. This radiation travels across the short, isolated gap between the connector heads and is captured by the sensor array (220) in the RX connector head (110b). The IC (230) in the RX head functions as a decoder, analyzing the captured sequence of frames. It reads the precise emission characteristics and position of each channel and translates this information back into the original digital data, which is then sent to the receiving computing device (302).

A significant innovation of the system is a built-in negotiation and calibration protocol.

Upon connection, the RX unit's IC initiates a handshake with the TX unit's IC. During this handshake, the RX unit queries the TX unit for its array capabilities, including but not limited to: maximum resolution, available wavelengths and intensities (e.g., color depth and matrix), and refresh rate. The RX unit uses this information to calibrate its decoding algorithms, creating a precise mapping of expected emission characteristics from the TX unit to the values registered by its own sensor. This self-calibration dramatically increases the precision of the system, reduces noise, and allows for interoperability between different hardware revisions or manufacturers.

Several physical embodiments and coupling configurations are contemplated, as shown in **FIG. 5**.

1. **Emitter Array Larger than Sensor Array (501):** The transmitter array has a larger surface area than the receiver sensor. This configuration can be useful in applications where some data redundancy is desired.

2. **Sensor Array Larger than Emitter Array (502):** The receiver sensor has a higher resolution and/or larger surface area than the transmitter array. This allows for techniques like oversampling, where multiple sensor elements capture a single emitter element, increasing precision and reducing errors.

3. **1:1 Aligned Coupling (503):** In a specialized embodiment for applications requiring maximum fidelity, such as in quantum computing, the emitter and sensor arrays are of identical size and resolution. The interlocking mechanism ensures near-perfect physical alignment, such that each emitter corresponds directly to a single sensor element. While mechanically complex and costly, this variant offers the highest possible data integrity.

To address the mechanical challenges of the 1:1 aligned coupling, a more robust and cost-effective variant is proposed, illustrated in **FIG. 4**. In this embodiment, instead of mapping data to a single physical emitter element (401), the encoder groups a block of physical emitters (e.g., a 10x10 or 100x100 grid) to form a single "channel element" (402). The entire block is instructed to emit with the same characteristics. This sacrifices some of the raw data density of the system but makes the alignment of the connector heads far less sensitive to minor physical imperfections, making it ideal for commercial datacenter applications where reliability and cost are paramount. The immense inherent bandwidth of the technology ensures that even with this reduction in density, the data transfer rates remain orders ofmagnitude above conventional solutions.

Claims

What is claimed is:

1. A data transmission system, comprising:

    - a transmitter unit including a two-dimensional array of individually addressable electromagnetic radiation emitters;
    - an encoder circuit configured to receive digital data and to generate a sequence of frames by mapping said digital data to the emission characteristics and spatial positions of the emitters in the array;
    - a receiver unit including an array of electromagnetic radiation sensors configured to capture the sequence of frames emitted by the transmitter unit;
    - a decoder circuit configured to generate a replica of the digital data by interpreting the emission characteristics and spatial positions of the emitters in the captured frames; and
    - a physical housing assembly configured to optically couple the transmitter unit and the receiver unit while shielding them from ambient radiation.

2. The system of claim 1, wherein the emission characteristics include at least one of wavelength or intensity.

3. The system of claim 1, wherein the electromagnetic radiation emitters and sensors operate in the visible light spectrum.

4. The system of claim 3, wherein the transmitter unit is a micro-display and the receiver unit is an image sensor, and wherein the emission characteristics are the color and brightness of pixels.

5. The system of claim 1, wherein the electromagnetic radiation emitters and sensors operate in a non-visible spectrum, such as infrared or ultraviolet.

6. The system of claim 1, wherein the data transmission bandwidth of the system is a direct function of the number of emitters in the array.

7. The system of claim 1, further comprising a communication protocol wherein the receiver unit is configured to query the transmitter unit to determine a set of emission capabilities and to calibrate the decoder circuit based on said capabilities.

8. The system of claim 1, wherein the encoder circuit is configured to form a "channel element" by assigning the same emission characteristics to a predefined block of adjacent emitters in the array.

9. A method for transmitting data, comprising the steps of:

    - receiving, at a transmitter, a stream of digital data;
    - encoding said data into a sequence of frames, wherein each frame comprises a plurality of emission channels, and wherein each channel is assigned an emission characteristic corresponding to a portion of the data;
    - emitting said sequence of frames from a two-dimensional array of emitters;
    - capturing said sequence of frames with an array of sensors optically coupled to the array of emitters;
    - decoding the captured frames by reading the emission characteristic and position of each channel to reconstruct the stream of digital data; and
    - outputting, from a receiver, the reconstructed stream of digital data.

10. The method of claim 9, wherein the emission characteristic is selected from the group consisting of: wavelength, intensity, color, and brightness.

11. The method of claim 9, further comprising the preliminary step of:

    - transmitting, from the transmitter to the receiver, a set of parameters describing the capabilities of the emitter array; and
    - calibrating, at the receiver, a decoding map based on said parameters.

Drawings

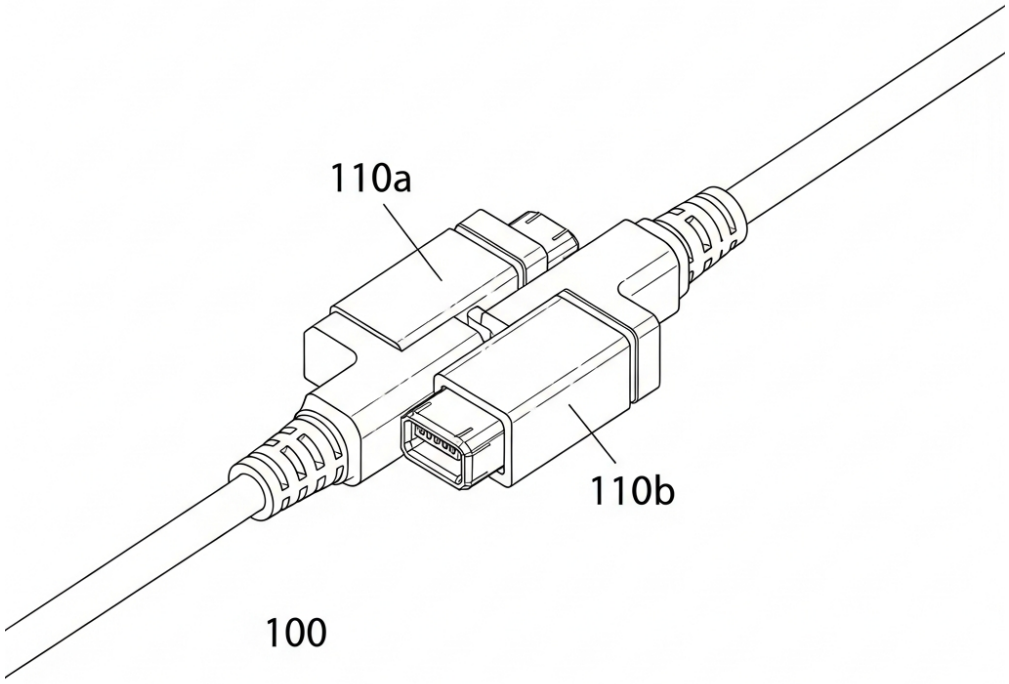


Figure 1: Isometric view of the assembled optical data cable.

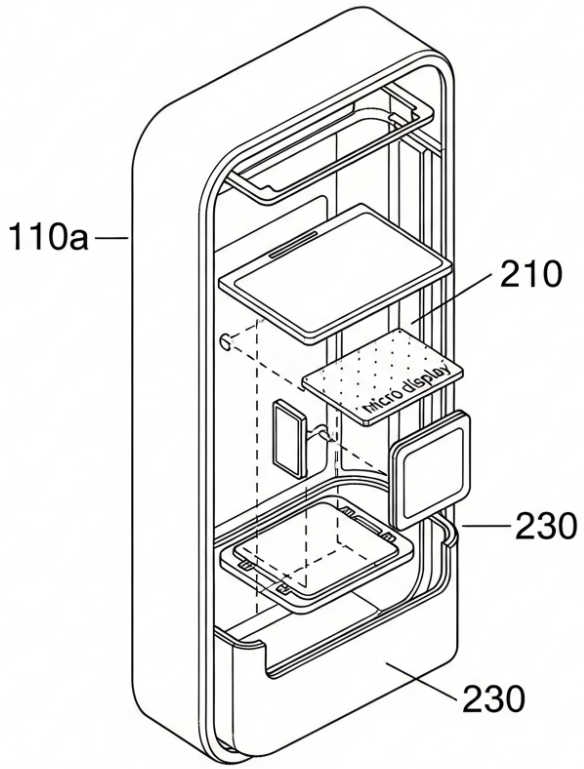


Figure 2: Exploded view of a connector head.

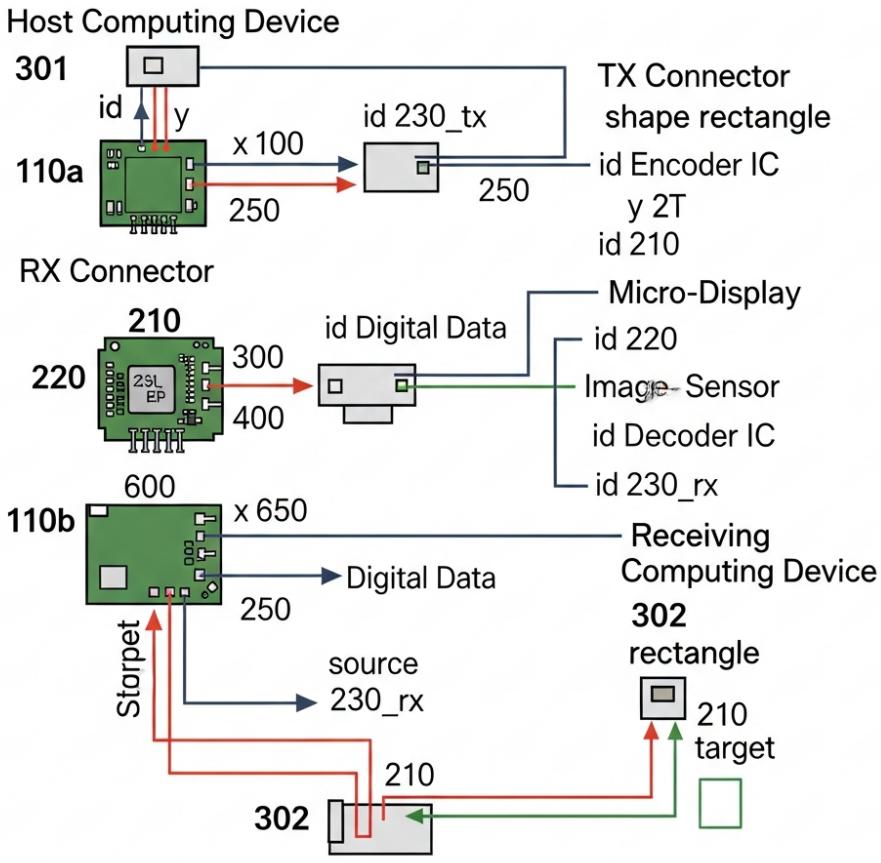


Figure 3: Block diagram of the data flow path.

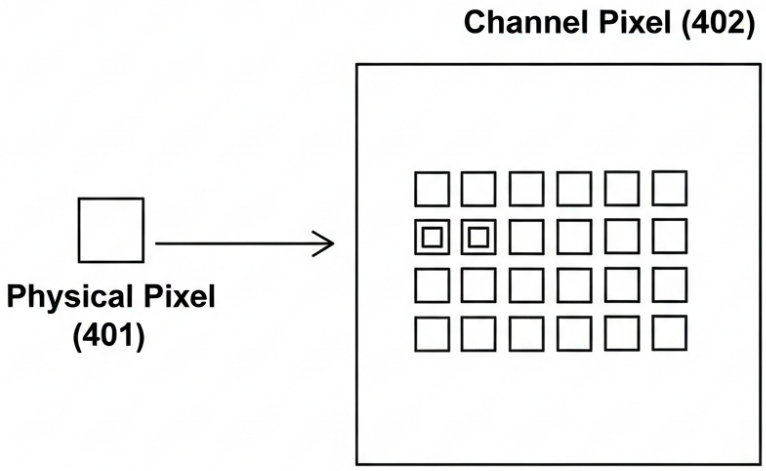


Figure 4: Conceptual diagram of a "channel pixel".

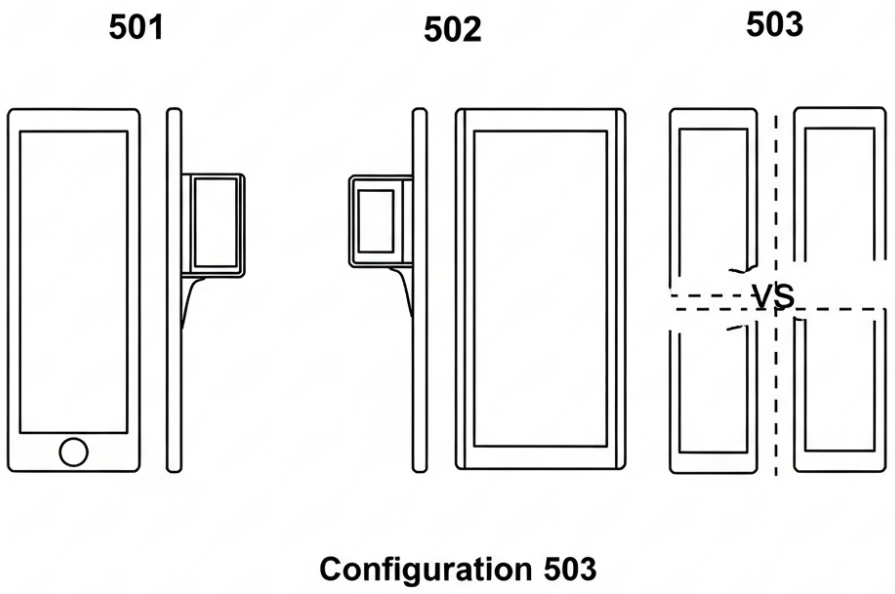


Figure 5: Physical coupling configurations.